

Integrated Stock Analysis and Portfolio Optimization: Tools for Individual Investors



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ABSTRACT

Stocks have become a globally popular investment instrument in the world. Both teenagers and adults are becoming more interested in technical understanding of stocks investment. In order to empower investors with technical understandings, this study presents an integrated approach for stock portfolio analysis, optimization, and backtesting tailored for individual investors. The project consists of three key components, firstly a stock search engine, secondly portfolio analysis module, and a backtesting system, all combined within an interactive GUI to enhance better user experience and accessibility.

The stock search engine allows users to retrieve real-time Indonesian stocks data. Providing visualizations like line and candlestick charts where users will get information about historical price patterns and also the company information, enhancing their understanding of price trends and supporting their investment decisions.

The portfolio analysis module employs Modern Portfolio Theory (MPT) to calculate optimal stock allocations by balancing return and risk, while staying with the user's budget constraint. With a combination of expected returns, covariance matrix, and knapsack adjustment, the model provides tailored asset allocations aimed at achieving a specified target return with minimized risk.

The backtesting system allows users to select certain stocks and a defined period to evaluate their performance over time using technical indicators such as SMA, EMA, RSI, etc, and several combinations. By applying these strategies to historical data, the system generates insights into key performance metrics, including Sharpe Ratio, drawdown, and volatility, specific to the selected period and stocks. This approach provides investors with a practical analysis of how chosen stocks would have performed, helping to inform future investment strategies and optimize portfolio management.

INTRODUCTION

Background and Motivation :

"The growing accessibility of stock markets and online trading platforms has led to increased participation from individual investors. However, managing investments effectively requires advanced tools to navigate market complexities."

In stock investment, analyzing stock performance over time is crucial for making decisions. Investors often need accessible tools to evaluate historical price movements and identify trends for quantitative analysis (American Economic Association, 2020), which can guide them in selecting promising stocks. One of our programs, Stock Search Engine, gives investors an overview of stocks' past performance and brief explanations that may help investors in identifying potential stocks.

Portfolio optimization is a fundamental concept that aims at maximizing returns while managing risk (Reilly et al, 2002). It is a great challenge for individual investors to create the most efficient capital use. The optimization in our programs helps investors to allocate their portfolio that aligns with their return objectives and risk tolerance. It combines user budgets and target returns, making it accessible to a range of investors, from conservative to more aggressive. By factoring in risk via the covariance matrix and using constraints like the budget, the program can effectively allocate assets to achieve the desired return with minimized risk.

In order to validate investment strategies, backtesting is one important step since it allows investors to simulate how a strategy would have performed in the past (Suhonnen et al, 2017). By allowing users to select particular stocks and time periods, our backtesting feature offers flexibility in evaluating different strategies and technical indicators (e.g., RSI, MACD). Those strategies differ in their buying and selling stocks behavior, resulting in different profit accumulation.

Problem :

There are several limitation in existing financial tool;

1. These tools are rarely available for Indonesian stocks.

2. Most stock analysis tools focus on a single aspect of analysis and rarely integrate search engines, portfolio optimization, and backtesting into a cohesive system.
3. Most tools use only one indicator for backtesting analysis, limiting the depth of strategy evaluation.
4. Existing optimization models often fail to account for constraints such as budget and risk, making them less practical for real-world use.

Improvement Proposed :

To overcome the existing problems, this study proposes a comprehensive tool that integrates stock analysis, portfolio optimization, and backtesting. Key innovations include:

1. **Portfolio Optimization Module:**
 - Uses Modern Portfolio Theory (MPT) to balance risk and return.
 - Introduces a knapsack adjustment to ensure practical allocations within budget and risk constraints.
 - Give recommendations based on the output.
2. **Backtesting Module:**
 - Find the best indicator's combination that may give the best return for each stocks
 - Find the best buy/sell strategies that will support the indicators to maximize the return.
 - Give prediction for the next day

Goal and Objectives :

1. Develop a tool that enables investors to make data-driven decisions by analyzing historical stock performance.
2. Optimize portfolio allocations within practical constraints to balance risk and return.
3. Validate and refine investment strategies using backtesting methods.

Tentative Result :

The proposed tool is expected to:

1. Provide investors with actionable insights into stock performance and market trends.
2. Generate realistic and optimized portfolio allocations that align with user-defined financial goals.
3. Validate strategies effectively using backtesting and also offering robust performance metrics such as Sharpe Ratio and net profit.
4. Enhance user confidence in making informed and practical investment decisions.

Benefit :

If successful, the proposed tool will:

1. Simplify the investment process for individual investors by integrating multiple functionalities into one system.
2. Enable practical, real-world portfolio construction optimization with real budget and risk constraints.
3. Improve investment strategy validation with backtesting method and actionable recommendations.
4. Support better risk management and decision-making through advanced analytics and visualizations.

Related Work

Portfolio Management has long been studied in stock finance, Modern Portfolio Theory has been used for the common theory and the foundation for balancing risk and return. MPT uses the covariance matrix to show the asset correlations, enabling diversifications to minimize the risk (Markowitz, 1952). Our project will implement MPT as part of the optimization process, we will extend it by incorporating the knapsack approach to handle practical constraints such as budget limits and expected returns. This approach will support the needs of individual investors to personalized their portfolios based on their risk tolerance and return goals. Previous studies ,

such as those by Vaezi et al in 2019 have explored similar constraint-based models, However, their approach still remains theoretical and only designed for institutional use. Our approach will advance accessibility for individual investors to dynamically adjust their budgets based on MPT and Knapsack within an GUI interactive program.

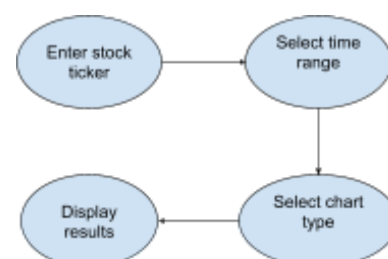
Backtesting is also a critical tool that investors often use for accessing the investment strategies as Bailey et al. (2015), suggest about overfitting for validating results. Fischer and Krauss (2018) examined the performance of various technical indicators, including RSI, MACD, and moving averages, showing that indicator effectiveness can vary significantly with market conditions. While many studies focus on evaluating individual indicators, our backtesting program enables users to test multiple indicators simultaneously, allowing them to identify the optimal combination of indicators and strategies for specific market scenarios. Our project provides flexibility, showing visual comparisons to find the best strategy. Flexibility merged with visual comparisons in our GUI and also overfitting test, differentiates our approach by giving investors a comprehensive toolkit for strategy evaluation and selection.

Methodology

1. Framework :

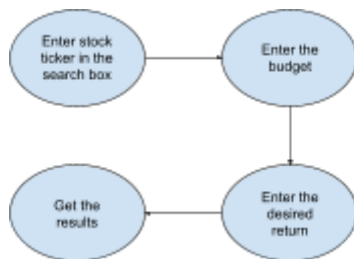
Stock Search and Analysis:

1. Input Ticker: Users enter the stock ticker in the search box (e.g., BBKA for Bank Central Asia).
2. Select Time Range: Choose from predefined time periods (e.g., 1 Year, 6 Months) for analyzing historical data.
3. Select Chart Type: Choose between a line chart or candlestick chart for visualizing stock trends.
4. Display Results: The tool retrieves and displays data, along with sector information, industry details, and historical price movements.



Portfolio Optimization:

1. Define Budget: Users input their budget
2. Desired Return: Enter the target return percentage, allowing the model to process the portfolio for a balanced risk-return profile.
3. Optimization Results: The tool calculates and displays optimized allocations based on Modern Portfolio Theory and Knapsack, along with expected returns and risk metrics.



Backtesting:

1. Select Stocks and Period: Choose specific stocks for strategy testing.
2. Apply Indicators: The tool applies technical indicators (e.g., SMA, RSI) and finds the best combination results and best strategy performance.
3. View Results: Users receive a summary of performance metrics (Sharpe Ratio)



2. Data Collection :

The data is gathered by using Yahoo Finance API from python library. The historical price data includes open, high, low, close prices, volume data and also explanation on each stock.

3. Sampling:

- Portfolio analysis, this program is testing using the last 1 year as the basis return to analyze the stocks allocation

- In case of a backtesting test, it analyzed the historical stocks data from the users input and time range.

4. Design:

- a. Phenomena studied
 - Stock prices and volatility Trends
 - Risk return relationships
 - Varies of investment strategies using backtesting method
- b. Key Elements
 - Stock analysis: Line charts, candlestick visualizations.
 - Portfolio optimization: Budget constraints, expected returns.
 - Strategy backtesting: Equity curve and metrics comparison.

5. Research Questions:

- How does incorporating knapsack adjustments improve the feasibility of allocations?
- Which technical indicator combinations provide the highest Sharpe Ratio?
- Which technical indicator combinations provide the best strategy?
- How do different strategies perform under backtesting?

6. Hypothesis :

- Knapsack can adjust the weight of budget limits and expected return to make portfolio allocations
- Combining indicators will improve strategy performance compared to using a single indicator.
- The backtesting program can analyze the selected stocks and time range and give an effective indicator signal.

7. Models

Portfolio Management

- Objective : maximize $\omega^T \mu$
- Budget Constraint :

$$\sum \omega_i \cdot P_i \leq B$$

P_i = price of stock i

B = budget

- Risk constraint :

$$\omega^T \sum \omega \leq R_{max}$$

R_{max} =
maximum allowable risk

- Non-negativity and full investment :

$$\sum \omega_i = 1$$

$$\omega_i \geq 0$$

- Knapsack Adjustment

$$L_i = \frac{\omega_i \cdot B}{P_i \cdot 100}$$

L_i = number of lots for stock i
100 = Lot size for each stock

Backtesting Analysis

- Objective : Maximizing Strategy Performance
- Performance metrics include:
 - Net Profit: Total returns from the strategy.
 - Sharpe Ratio: Measures risk-adjusted returns.
 - Profit Factor: Ratio of total profit to total loss.
- Constraint :
 - Max Drawdown should be less than 20
 - Sharpe Ratio should be more than 1

8. Indicators Used

8.1. Simple Moving Average (SMA)

Definition:

- The SMA is the average of a stock's closing prices over a specific period of time. It smooths out price data to identify the direction of the trend.

Formula:

$$SMA_t = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i}$$

P_{t-i} = closing price at day $t - i$

n = number of periods

Usage:

- Buy Signal: When a short-term SMA (e.g., 10-day) crosses above a long-term SMA (e.g., 50-day).
- Sell Signal: When a short-term SMA crosses below a long-term SMA.

8.2. Exponential Moving Average (EMA)

Definition:

EMA gives more weight to recent prices, making it more responsive to current price movements compared to the SMA.

Formula:

$$EMA_t = P_t \cdot k + EMA_{t-1} \cdot (1 - k)$$

EMA_t = closing price at day $t - i$

P_t = current price

$k = \frac{2}{n+1}$; smoothing factor

n = number of periods

Usage:

- Buy Signal: When a shorter EMA (e.g., 12-day) crosses above a longer EMA (e.g., 26-day).
- Sell Signal: When a shorter EMA crosses below a longer EMA.

8.3. Relative Strength Index (RSI)

Definition:

- The RSI measures the speed and magnitude of price changes to identify overbought or oversold conditions.

Formula:

$$RSI = 100 - \frac{100}{1 + RS}$$

$RS = \frac{\text{Average Gain}}{\text{Average Loss}}$; ratio of average gains to average losses over a specific period

Usage:

- Overbought: $RSI > 70$ (potential sell signal).
- Oversold: $RSI < 30$ (potential buy signal).

8.4. Moving Average Convergence Divergence (MACD)

Definition:

- The MACD tracks the relationship between two EMAs (short-term and long-term) to identify trend changes.

Formula :

1. MACD Line : Difference between the 12-day EMA and the 26-day

$$MACD = EMA_{12} - EMA_{26}$$

2. Signal Line : 9 Day EMA of the MACD line
3. Histogram : Difference between the MACD Line and the signal line

$$\text{Histogram} = MACD - \text{Signal Line}$$

Usage:

- Buy Signal: When the MACD line crosses above the Signal line.

- Sell Signal: When the MACD line crosses below the Signal line.

8.5. Bollinger Bands

Definition:

- Bollinger Bands consist of three lines: a simple moving average (SMA) and two bands representing standard deviations of price above and below the SMA.

Formula:

Middle band : SMA of closing Prices

$$\text{Upper Band} = SMA + (k \cdot \sigma)$$

$$\text{Lower Band} = SMA - (k \cdot \sigma)$$

Where:

σ = Standard deviation of the price

k = Multiplier, typically 2

Usage:

- Buy Signal: When the price touches or moves below the Lower Band.
- Sell Signal: When the price touches or moves above the Upper Band.

8.6. On-Balance Volume (OBV)

Definition:

- OBV measures cumulative buying and selling pressure based on volume. It adds volume on up days and subtracts volume on down days.

Formula:

$$OBV_t = -OBV_{t-1} + Volume_t, \text{ if } P_t > P_{t-1}$$

$$OBV_{t-1} - Volume_t, \text{ if } P_t < P_{t-1}$$

$$OBV_{t-1}, \text{ if } P_t = P_{t-1}$$

Where :

P_t = closing price at time t

$Volume_t$ = trading volume at time t

Usage:

- Trend Confirmation: Rising OBV confirms an uptrend, while falling OBV confirms a downtrend.
- Divergence: If OBV moves in the opposite direction of the price, it signals a potential reversal.

9. Strategy

- Basic Strategy

Rule : Buy when a buy signal is triggered, and sell on the next sell signal.

Implementation: Track buy/sell signals in the backtest function and execute trades accordingly.

Use Case: Suitable for short-term trading where quick response to signals is desired.

- Accumulation Strategy

Rule: Continue buying on each buy signal without selling until a sell signal occurs, then sell all positions.

Implementation: Maintain a "position" variable to track accumulated buys. When a sell signal is triggered, close all positions.

Use Case: Useful for bullish trends where you want to

maximize accumulation and then exit fully on a downward trend.

- Profit locking strategy

Rule: Only sell if the current price is higher than the last buy price, ensuring a profit on each trade.

Implementation: Track the last buy price, and only execute a sell if the current price exceeds the buy price.

Use Case: Useful in sideways markets to avoid losses on each sell; it also controls risk by selling only on profitable signals.

10. Input Variables

- Stock tickers.
- Timeframe
- Budget and target return.

11. Control Variables

For backtesting analysis

- Selected strategies (e.g., Basic, Accumulation, Profit Locking).
- Technical indicators (e.g., RSI, MACD, Bollinger Bands, EMA, SMA).

12. Policy Variables

For Portfolio Analysis

- Target return threshold : Defines the minimum acceptable return for the portfolio, guiding the optimization process to prioritize strategies that meet or exceed this threshold.

13. Output

- Final Portfolio Weights: Indicates the percentage allocation for each stock in the optimized portfolio after considering constraints.
- Combined Trading Signals: Show signals from the multi-indicator signal matrix, to guide users for buy/sell

decisions during backtesting and also the equity curves.

14. Performance Indices

- Sharpe Ratio: Measures risk-adjusted returns.

$$\text{Sharpe Ratio} = \frac{\text{Mean Return} - \text{Risk Free Rate}}{\text{Standard Deviation of Returns}}$$

- Net Profit: Total monetary gain or loss over the backtesting period.
- Maximum Drawdown: The largest drop from a peak to a trough in the equity curve, indicating portfolio risk.

$$\text{Max Drawdown} = \max \left(\frac{\text{Peak Value} - \text{Trough Value}}{\text{Peak Value}} \right)$$

15. Parameters

- Risk-Free Rate (4.8%): Used in the Sharpe Ratio calculation to benchmark portfolio performance against a risk-free investment.
- Volatility Thresholds for Portfolio Risk: Constraints on the portfolio's variance to limit exposure to high-risk assets.

16. Search/ Optimization

- Find the best Stocks allocation that minimizes risk with budget constraint.
- Find the best Indicator's combination and strategy to get the best return.

Implementation and Result

1. Search Engine

Stock Search Portfolio Optimization

Enter Stock Ticker (e.g., TSLA, BRCA)

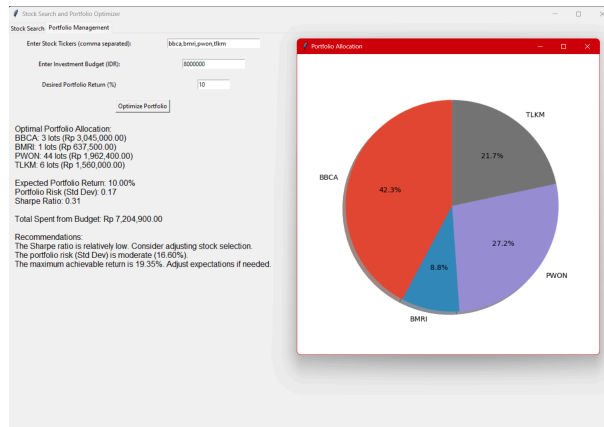
Select Time Range Select Chart Type

Name: Bank Central Asia Tbk
Sector: Financial Services
Industry: Banks - Regional
Current Price: N/A
Previous Close: 10050.0



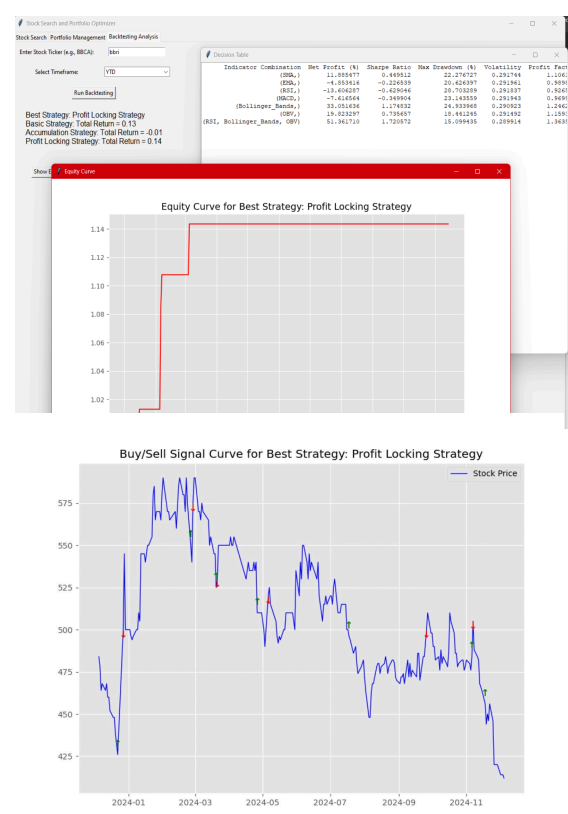
- Visualization: This section will generate a chart(line/candlestick) that shows price trends over the selected time frame.
- Insight : This visualization and information will give investors quick insight into the stock's performance:
 - Stock Information : Key details, such as the company's sector and closing price will provide investors some quick perspective on the stock's performance potential.
 - Trend identification : The line and candlestick chart will help investors identify whether the stock has been in an upward (bullish) or downward (bearish) trend, allowing for better timing of investment decisions based on past performance.

2. Portfolio Optimization



- Interpretation : Using two main theories, Modern Portfolio Theory (MPT) and knapsack, it creates a portfolio allocation based on the budget and risk constraint. The adjustment allows the user to see a feasible allocation that can be directly implemented in the market.
- Insight : The portfolio allocation created will support the users to create an optimized investment portfolio that aligns with their financial goals, budget, and risk tolerance . Key benefits:
 - Optimized Investment Decisions: By balancing risk and return, it will help users to make an optimal decision.
 - Clear and Actionable Recommendations: The tool provides specific recommendations, such as adjusting stock selections to improve the Sharpe Ratio or considering alternative stocks if the desired return is unrealistic. This guidance helps users to allocate their portfolio better and align with their investment goals.
 - Clear Allocations: With a visual breakdown of allocations, users can easily see how their capital is distributed across different stocks, helping them achieve a diversified portfolio with calculated risk.

3. Backtesting



Interpretation:

The output provides a depth analysis of various indicator combinations and trading strategies applied to a selected stock over a specified period. By comparing metrics such as net profit, Sharpe Ratio, maximum drawdown, volatility, and profit factor, the tool identifies the best-performing indicators and strategies. Additionally it will also show the price graph where the program gives signals on the historical prices. This allows users to understand which approaches offer optimal returns while managing risk effectively.

Insight:

The backtesting analysis supports users in selecting the most effective strategy and indicators for maximizing returns and managing risk. Key benefits include:

- Effective Indicator Combination: The decision table reveals the most profitable and stable indicator combinations, such as SMA, RSI, and Bollinger_Bands, which balance high Sharpe Ratio and moderate drawdown. This insight helps users choose indicators that align with their risk tolerance and performance goals.

- **Informed Strategy Selection:** The tool evaluates multiple strategies (e.g., Basic, Accumulation, Profit Locking) and identifies the best-performing one based on total return. In this case, the Accumulation Strategy shows the highest return, indicating it as the optimal choice for long-term growth.
- **Clear Performance Visualization:** The equity curve for the best strategy provides a visual representation of portfolio growth over time, showing a consistent upward trend. This allows users to assess the stability and growth potential of the chosen strategy, helping them make decisions aligned with their investment horizon.
- **Short-Term Decision Guidance:** The prediction for the next day (buy/sell) offers actionable advice, guiding users on potential exit points based on the latest data. This real-time insight helps users respond to market changes and make tactical adjustments to their portfolio,

Conclusion and Recommendation

Our program successfully fulfilled the primary objectives of developing an integrated stock analysis tool by implementing Operation research and also Linear Algebra lessons. Here are the key conclusions :

1. **Search engine**
This tool provides a simple program to retrieve and show historical price data. By enabling users to evaluate trends, stock information, it will support the user's investment decisions.
2. **Portfolio Optimization**
Using Knapsack and Modern Portfolio Theory (MPT), the tool has succeeded in optimizing asset allocations. It balances expected return with risk based on the user's constraint such as budget and target return. This optimization allows users to adapt practical and actionable portfolio management.
3. **Backtesting Analysis**
This tool evaluates the best indicators and strategies by evaluating their performance over historical periods. The ability to test and find different technical indicators and strategies based on the best indicators provide effective strategies.

Based on the results, we proposed the following recommendations for further study and improvement:

1. **Check the logic to avoid overfitting**
Current approach evaluates historical data by combining some technical indicators and 3 strategies that may risk overfitting to past trends, which may not be generalized for the future implementation.

Recommendation : Use cross-validation techniques or forward testing to ensure the strategies perform well on sample data, this will reduce the risk of overfitting and provide better evaluation.

2. **Incorporating Transaction cost**
Our current program does not apply the real-world trading costs, such as brokerage fees and taxes, which may lead to significant impact on portfolio performance.

Recommendation : Add transaction cost calculations for portfolio management and also backtesting analysis. It will lead to more realistic performance metrics and align better with the real-world scenarios.

3. **Expanding Indicator choices**
The program currently only supports popular technical indicators such as RSI, MACD, SMA, EMA, Bollinger Bands.

Recommendation : Include more advanced technical indicators such as fibonacci, stochastic or other volume-based indicators. These additions would allow users to evaluate more variative strategies.

4. **Expanding strategy choices**
The tool currently provides Accumulation, basic and also profit locking strategies, the

addition of different approaches may improve its utility.

Recommendation : Explore more strategies such as momentum-based trading, apply take profit and stop loss , etc. Each strategy will give different insight and result on the portfolio.

5. Incorporating Machine Learning

Applying machine learning could enhance predictive accuracy and strategy, enabling it to self-detect the non-linear relationship on the data

Recommendation :
Implement machine learning models for stock price prediction
Train the models to carefully selected features to avoid overfitting and ensure generalization to future data.

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